

Final Projects and Frontiers of QML

EVA: Quantum Machine Learning – Lecture 14

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1. Student project presentations (30–40 min)
2. Frontiers: what is happening now in QML research
3. Dequantization and honest classical baselines
4. The Swiss quantum ecosystem
5. Course recap: what is proven, what is promising, what is hype
6. Open research problems and next steps

Part 1. Project presentations

Presentation format and discussion rubric

Each team presents for 5–7 minutes, followed by 2–3 minutes of questions.

The discussion follows a four-point rubric:

Point	Question
Problem statement	Is the task well-defined? Is the data or problem clearly described?
Baseline	Was a classical or simpler quantum baseline established?
Evaluation	Are the metrics appropriate? Are the comparisons fair?
Limitations	Does the team acknowledge what their approach cannot do?

The rubric applies equally to papers in the literature: a project that honestly identifies its own limitations is more credible than one that overstates its results.

Part 2. Frontiers of QML

Quantum transformers and attention

Quantum Doubly Stochastic Transformer (QDST): Born et al. (NeurIPS 2025) showed that unitary matrices produce doubly stochastic attention weights ($|U_{ij}|^2$), preventing attention collapse. Classical simulation experiments on small NLP tasks showed competitive performance.

Open questions:

- does the doubly stochastic prior actually improve generalization, or is it equivalent to a classical regularizer?
- can the quantum attention be implemented efficiently on hardware for sequence lengths of practical interest ($n > 20$)?
- can the attention weights be made *trained* (not fixed random)?

Orthogonal attention: Classical methods can also produce doubly stochastic attention via the Sinkhorn algorithm or Householder reflections. The quantum advantage (if any) must be demonstrated beyond structural similarity.

QPINNs for scientific computing

Recent experiments with Quantum Physics-Informed Neural Networks on physically relevant problems:

Equation	Approach	Status
Harmonic oscillator ODE	1–2 qubit QPINN	Proof of concept; matched classical PINN
1D heat equation	4-qubit QPINN, hybrid	Competitive with classical at ≤ 10 parameters
Maxwell equations	Hybrid QPINN (TU Delft group)	Early results; circuit depth limits frequency
Navier-Stokes (turbulent)	Classical PINN superior so far	QML not yet competitive; parameter explosion

The bottleneck is circuit expressibility: representing high-frequency components of PDE solutions requires deep circuits, which push into the noise-limited regime.

Quantum reservoir computing at scale

Key recent results:

IBM 156-qubit QRC (2025): Linear readout on Pauli measurements from a 156-qubit Heron processor achieves NMSE below 0.01 on the Lorenz-63 attractor. Classical simulation is intractable above 50 qubits for this circuit structure.

QuaRK kernel (2025): Reservoir-induced kernel outperforms fidelity kernels on time-series benchmarks at $\mathcal{O}(N \cdot m)$ evaluation cost rather than $\mathcal{O}(N^2)$.

Noise-induced QRC (IBM, 2023): Calibrated hardware noise improves Mackey-Glass prediction accuracy compared to noiseless simulation – the first case where hardware imperfection is a feature, not a bug.

These results suggest QRC is the most hardware-ready QML approach today: no circuit training needed, and the noise is exploitable.

Quantum ML for quantum data

The strongest theoretical case for QML is when the data itself is quantum: outputs of quantum experiments, molecular ground states, quantum error syndromes.

Quantum chemistry: Predicting molecular energies, reaction barriers, or spectral properties directly from quantum circuits that encode the molecular Hamiltonian. VQE and ADAPT-VQE are leading approaches. Current 50–100 qubit circuits with error mitigation match coupled-cluster (CCSD) accuracy for small molecules (< 20 atoms).

Materials science: Phase classification of quantum materials from their ground states (QCNN application, Cong et al. 2019); disorder and topological invariants.

Quantum error correction: Decoding surface code syndromes with a ML model trained on quantum hardware data. Recent work from Google and Microsoft shows classical NNs work well here; whether QML decoders offer advantages is open.

Part 3. Dequantization and honest baselines

Dequantization: Tang's result

Ewin Tang (2019) showed that several quantum algorithms claimed to offer exponential speedups for ML tasks have classical analogs with polynomial speedup.

The key technique: **quantum-inspired sampling**. Classical algorithms with ℓ^2 -norm sampling access to matrices (a classical data structure that supports importance sampling efficiently) can match the asymptotic scaling of quantum recommendation systems, PCA, and linear regression – previously thought to require quantum hardware.

Implications:

- proofs of exponential quantum advantage must be scrutinized carefully
- the speedup often comes from assuming quantum RAM (QRAM), not from quantum operations per se
- QRAM itself is a physical assumption that has no demonstrated hardware realization

Bottom line: if a classical algorithm with the same data access matches the quantum algorithm, the “quantum” advantage is not from quantum mechanics.

Tensor networks as classical competitors

Many quantum circuits for 1D systems (e.g. brickwork circuits, QCNN with 1D structure) can be exactly simulated by classical **tensor networks** with polynomial cost.

Matrix product states (MPS): for 1D quantum states with limited entanglement (bond dimension χ), MPS provides exact representation in $\mathcal{O}(\chi^2 n)$ memory.

Consequence for QML:

- any QCNN, QRC, or variational circuit on 1D topology with moderate depth may be simulatable by an MPS-based classical ML model
- the circuit must generate *long-range entanglement* to be classically hard
- current NISQ circuits rarely produce this level of entanglement (noise breaks it)

A rigorous comparison should always include an MPS-based baseline alongside the classical neural network baseline.

When to take a QML advantage claim seriously

A credible advantage claim should satisfy:

1. **Task is well-defined:** exact mathematical problem statement, not a vague benchmark.
2. **Classical baselines are fair:** comparison includes well-tuned classical ML, MPS/tensor networks, and QRAM-free classical analogs.
3. **The advantage is in the right resource:** time, sample complexity, or parameter count – not an unfair accounting trick.
4. **The data model is honest:** if QRAM is assumed, the cost of building QRAM is counted.
5. **Hardware noise is accounted for:** the advantage survives realistic error rates; it is not a simulation-only result.

Very few published QML results satisfy all five points simultaneously. This is not a reason for pessimism – it is a research opportunity.

Part 4. The Swiss quantum ecosystem

Swiss quantum landscape

Switzerland has a growing and well-funded quantum ecosystem:

Organization	Activity
ETH Zürich	Superconducting qubit research (D-Wave partnership); quantum adv
EPF Lausanne (EPFL)	Topological qubits; quantum photonics; quantum networks
University of Basel	Ion trap quantum computing; quantum simulation
QC2 (Basel + QuantumBasel)	Applied QML research, hardware-agnostic approaches, industry partn
QuantumBasel	Industry hub; D-Wave quantum annealer; applied optimization
ZHAW	QML education and applied research (this course!)
IBM Zurich Research	Quantum hardware; error mitigation; QML algorithms

Career paths in QML

The QML field spans academia and industry with distinct profiles:

Academia: Focus on proving rigorous complexity results, demonstrating novel architectures, and characterizing when quantum ML has provable advantages. Strong intersection with quantum physics, theoretical computer science, and statistics. Key venues: NeurIPS, ICML, Physical Review X, Nature journals.

Startups: Quantum chemistry (QunaSys, Zapata), optimization (1QBit, D-Wave), finance (Multiverse), simulation (Quintuum, IonQ). Require hybrid skills: quantum algorithms + software engineering.

Large industry (IBM, Google, Microsoft): Hardware development, error correction, algorithm teams. Entry through PhD or strong research track record.

Classical ML with quantum awareness: Understanding dequantization, tensor networks, and quantum-inspired algorithms is increasingly relevant for classical ML practitioners working on scientific problems.

Part 5. Course recap

QML: what is proven

Results with rigorous theoretical support:

- **Exponential feature space:** n qubits can represent a state in $\mathbb{C}^{2^n}$; the inner product (quantum kernel) is classically expensive to evaluate exactly for generic encodings.
- **Parameter-shift rule:** exact gradients of PQC expectations with 2 circuit calls per parameter.
- **Barren plateaus:** gradient variance decays as $\mathcal{O}(2^{-n})$ for random global-cost circuits (McClean et al. 2018).
- **Local cost avoidance:** QCNN and local observables provably avoid exponentially small gradients at polynomial depth.
- **QRC fading memory:** measurement-induced fading memory is necessary and sufficient for temporal processing.
- **Dequantization:** several “quantum speedups” in linear algebra have polynomial classical analogs (Tang 2019).

QML: what is promising

Results with empirical support, or theoretical support under specific assumptions:

- **Quantum kernels for structured data:** quantum kernels can capture correlations not represented by standard classical kernels; demonstrated on synthetic datasets with specific geometric structure.
- **QRC at scale:** IBM 156-qubit experiments show NMSE improvement with qubit count; classical simulation is hard above 50 qubits.
- **Quantum utility:** 127-qubit Ising simulation as a useful computational tool even without fault tolerance.
- **QPINNs for low-dimensional ODEs:** competitive with classical PINNs at matched parameter count for smooth, low-frequency solutions.
- **Noise as a resource in QRC:** calibrated noise improves prediction accuracy in reservoir computing experiments.

QML: what is hype

Claims that are overstated or unsubstantiated at this stage:

- **“Quantum ML will revolutionize AI.”** No rigorous proof of practical advantage on real ML benchmarks exists.
- **Exponential speedup for generic data.** Most speedups assume QRAM or structured input that does not arise in standard ML pipelines.
- **QNNs outperform classical NNs due to more parameters.** The parameter-count fallacy: a 4-qubit circuit has 8–20 parameters, far fewer than a competitive classical model.
- **Quantum computing solves NP-hard problems.** QAOA offers a constant-factor speedup at best for many problem instances; NP-hardness is not broken.
- **Noise is always a problem to be fixed.** Noise is sometimes exploitable (QRC), but this is not a general statement.

Decision framework: when to use quantum

A practical guide for choosing between quantum and classical approaches:

Condition	Recommendation
Data is classical, small dataset	Start with logistic regression; quantum unlikely to help.
Data is classical, structured (images, time series)	Try quanvolution or QRC as feature extractor; compare with classical CNN / LSTM.
Data is quantum (molecular, materials)	Strong candidate for QML; VQE, QCNN on quantum-native input.
Combinatorial optimization, small instances	QAOA ($p \leq 3$) as a tool, compare with simulated annealing.
Time series, chaotic dynamics	QRC with linear readout; check against classical ESN first.
Physics-informed	QPINN for low-dimensional ODEs; classical

Part 6. Open problems and next steps

Open research problems

1. **Rigorous classical simulation bounds:** For which circuit families and qubit counts does classical simulation become intractable? The moving boundary between quantum and classically tractable is the central open problem.
2. **Quantum advantage on naturally occurring ML tasks:** No real-world dataset + task combination has been proven to benefit from quantum circuits. What properties of a dataset would make this likely?
3. **Expressibility vs generalization:** Quantum circuits can be highly expressive; how does this translate to generalization bounds? The VC-dimension of PQCs is not well-characterized.
4. **Fault-tolerant QML:** Once error correction is available, which quantum ML algorithms become practical? What changes about the barren plateau story with logical qubits?
5. **Quantum data for classical models:** Can outcomes of quantum experiments (chemistry, materials) be fed into classical transformers with quantum-aware feature engineering?

Recommended next steps

To go deeper in QML after this course:

Theory: Schuld and Petruccione, *Machine Learning with Quantum Computers* (Springer, 2021). Cerezo et al., *Variational quantum algorithms* (Nat. Rev. Phys., 2021). Read original barren-plateau papers (McClean 2018, Cerezo 2021).

Practice: PennyLane demos: <https://pennylane.ai/qml/demos/>. IBM Quantum Learning: <https://learning.quantum.ibm.com/>. Work through the Qiskit textbook QML chapter.

Research track: Monitor NeurIPS, ICML, and Physical Review X for new results. arXiv preprint server: quant-ph section, search “quantum machine learning.” Join the PennyLane Forum or Qiskit Slack for community discussion.

Hardware access: IBM Quantum Network provides free cloud access for researchers and educators. Apply via <https://quantum.ibm.com/>.

Summary

Topic	Key takeaway
Project rubric	Problem, baseline, evaluation, limitations
Frontiers	QDST, QPINNs, QRC at scale, quantum-native data
Dequantization	Classical analogs exist for many “quantum speedups”
Honest baselines	Include MPS, QRAM-free classical algorithms
Swiss ecosystem	ETH, EPFL, Basel, IBM Zurich, QuantumBasel, ZHAW
What is proven	Barren plateaus, parameter-shift, fading memory, dequantization
What is promising	QRC at scale, quantum utility, quantum kernels for specific data
What is hype	General AI revolution, NP-hard solution, QRAM-free speedups
Decision framework	Match architecture to data type; classical baseline first

Quantum machine learning is a field that rewards skepticism and benefits from careful experimentation. The most valuable skill is knowing when not to use a quantum approach.